Conditional Expectation

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1 Setup

Throughout, $(\Omega, \mathcal{F}, P)$ is a probability space and all random variables (RVs) are real-valued unless otherwise stated. We write $L^p = L^p(\Omega, \mathcal{F}, P)$ for the usual Lebesgue spaces, and for a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ we write $L^p(\mathcal{G})$ for the closed subspace of $\mathcal{G}$-measurable representatives.

2 Conditional expectation given a random variable

Let X, Y be RVs with joint density $f_{X,Y}$ (the discrete case is analogous, replacing densities by PMFs and integrals by sums). Whenever $f_Y(y) > 0$, define the *conditional density of X given $Y = y$* by

$$f_{X|Y}(x \mid y) \;=\; \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

Definition 1 (Conditional expectation given $Y = y$)**.** For each y with $f_Y(y) > 0$,

$$E[X \mid Y = y] \;=\; \int_{\mathbb{R}} x \, f_{X|Y}(x \mid y) \, dx,$$

assuming the integral is well-defined. The map $g : y \mapsto E[X \mid Y = y]$ is a Borel function; the random variable

$$E[X \mid Y] \;:=\; g(Y)$$

is called the *conditional expectation of X given Y*.

3 Conditional expectation given a σ-algebra

Definition 2 (Conditional expectation given $\mathcal{G}$)**.** Let $X \in L^1$ and $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-algebra. A random variable Z is a *version* of $E[X \mid \mathcal{G}]$ if

(i) Z is $\mathcal{G}$-measurable;

(ii) $\displaystyle\int_A Z \, dP = \int_A X \, dP$ for every $A \in \mathcal{G}$.

Existence and a.s.-uniqueness of Z follow from the Radon–Nikodym theorem applied to the signed measure $\nu(A) := \int_A X \, dP$ on $\mathcal{G}$, which is absolutely continuous with respect to the restriction $P|_{\mathcal{G}}$. The earlier definition is recovered by taking $\mathcal{G} = \sigma(Y)$: by the Doob–Dynkin lemma, every $\sigma(Y)$-measurable RV is of the form $g(Y)$, and one checks that $g(y) = E[X \mid Y = y]$ satisfies (i)–(ii).

4 The tower property

Theorem 3 (Tower property)**.** *Let $X \in L^1$ and let $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$ be sub-σ-algebras. Then almost surely*

$$E\big[\, E[X \mid \mathcal{G}] \;\big|\; \mathcal{H} \,\big] \;=\; E[X \mid \mathcal{H}].$$

In particular, taking $\mathcal{H} = \{\emptyset, \Omega\}$ gives the unconditional identity

$$E\big[\, E[X \mid \mathcal{G}] \,\big] \;=\; E[X].$$

Proof. Write $Z := E[X \mid \mathcal{G}]$ and let W be any version of $E[Z \mid \mathcal{H}]$; we show W is also a version of $E[X \mid \mathcal{H}]$.

(i) $\mathcal{H}$-measurability. By the defining property of $E[Z \mid \mathcal{H}]$, the RV W is $\mathcal{H}$-measurable.

(ii) Defining integral identity. Fix any $A \in \mathcal{H}$. Because $\mathcal{H} \subseteq \mathcal{G}$, we have $A \in \mathcal{G}$ as well. By the definition of $W = E[Z \mid \mathcal{H}]$,

$$\int_A W \, dP \;=\; \int_A Z \, dP. \tag{$*$}$$

By the definition of $Z = E[X \mid \mathcal{G}]$, applied to the set $A \in \mathcal{G}$,

$$\int_A Z \, dP \;=\; \int_A X \, dP. \tag{$**$}$$

Chaining $(*)$ and $(**)$ gives

$$\int_A W \, dP \;=\; \int_A X \, dP \qquad \text{for every } A \in \mathcal{H}.$$

Combined with (i), this is precisely the defining property of $E[X \mid \mathcal{H}]$. By a.s.-uniqueness of conditional expectations, $W = E[X \mid \mathcal{H}]$ a.s.

The unconditional form is the special case $\mathcal{H} = \{\emptyset, \Omega\}$, for which $E[\,\cdot \mid \mathcal{H}]$ collapses to the ordinary expectation. □

Proposition 4 (L^2 projection)**.** *If $X \in L^2$, then $E[X \mid \mathcal{G}]$ is the orthogonal projection of X onto $L^2(\mathcal{G})$. Equivalently,*

$$E[X \mid \mathcal{G}] \;=\; \arg\min_{Z \in L^2(\mathcal{G})} E\big[(X - Z)^2\big]\,.$$

Example 5 (Sum of two dice)**.** Let Y and W be independent and uniform on $\{1, \ldots, 6\}$, and let $X = Y + W$. Then for $y \in \{1, \ldots, 6\}$,

$$E[X \mid Y = y] \;=\; y + E[W] \;=\; y + 3.5,$$

so $E[X \mid Y] = Y + 3.5$. The tower property gives

$$E[E[X \mid Y]] \;=\; E[Y] + 3.5 \;=\; 3.5 + 3.5 \;=\; 7 \;=\; E[X]. \quad \square$$